Colloquium Study Syllabus: Concepts Shared by All Predictors

Thesis: A Unified Benchmark of Grokking Predictors in Neural Networks · Common concepts (not predictor-specific) · Prepared 5 November 2026

This syllabus covers only the ideas that come up in every predictor (or nearly every one). Predictor-specific maths (KSG, O-information, HTSR α, commutators, PCA details) is left for later. Each topic gives: what it is in one line, why it matters in your thesis, a worked example with your own numbers, self-check questions, and a likely viva question.

Where it appears: `AGENTS.md` (predictor order and the Nanda gate), `context.md` (frozen-protocol section, 29–30 Sep 2026; criterion validation), `06_Code/scripts/analyze_nanda_unified.py` (`evaluate_frozen_protocol`), `06_Code/configs/nanda_unified.yaml`, `06_Code/scripts/run_nanda_benchmark.py` (checkpoint grid), and the attack-point documents in `11_Predictor_Issues_and_Lacunas`.

Suggested study order and time

Module	Topic	Time
1	The grokking phenomenon	1.5 h
2	Your training setup	2 h
3	Measuring the outcome: grok epoch	1 h
4	What a predictor is: event, lead, relative lead	1.5 h
5	Spearman's ρ and small samples	2 h
6	The frozen protocol and the PASS rule	1.5 h
7	Research honesty: bias, controls, negative results	2 h

Total about 11–12 hours. Do Modules 4–6 twice; they are what every predictor verdict rests on.

Module 1: The grokking phenomenon (3 topics)

1.1 Grokking

What it is: A network first scores perfectly on its training data, then much later suddenly scores well on unseen data too.

Why it matters: It's the event your whole benchmark tries to predict. Every predictor is judged by whether it signals *before* this jump.

Worked example: In your Nanda-Unified runs, training accuracy hits 100% within a few hundred epochs, but test accuracy only passes 0.9 at epochs 6,921 to 23,747 depending on the seed.

Self-check: (1) Why is grokking surprising? (2) What two curves do you need to see it?

Likely viva question: "Define grokking precisely. How is it different from ordinary slow learning?"

1.2 Memorisation vs generalisation

What it is: Memorisation means getting the training examples right by storing them; generalisation means getting new examples right because the rule was learned.

Why it matters: Grokking is the switch from the first to the second. Several of your predictors (e.g. L2) react to early memorisation instead of to that switch, which is an attack point.

Worked example: The L2 crossover fires at epochs 113–137, which is the memorisation phase, about 100 times earlier than grok.

Self-check: At epoch 1,000, is a typical run memorising or generalising? How do you know?

Likely viva question: "Is your signal detecting the switch to generalisation, or just the end of memorisation?"

1.3 Circuit formation and cleanup (Nanda's three phases)

What it is: Nanda et al. describe three phases: memorisation, circuit formation (the general algorithm is built quietly), and cleanup (the memorised parts are removed, and test accuracy jumps).

Why it matters: It explains why a hidden signal *could* exist before grok, which is the premise of the benchmark.

Worked example: Nanda report that $\sum w^2$ falls smoothly during circuit formation and sharply during cleanup, at the same time as grokking.

Self-check: Which phase would a good early-warning signal need to detect?

Likely viva question: "Nanda says the transition can't be predicted ahead of time. Why did you expect it could?"

Module 2: Your training setup (6 topics)

2.1 Modular addition task

What it is: The model learns $(a + b)$ mod p for all pairs $a, b \in \{0, \ldots, p-1\}$.

Why it matters: It's the standard grokking task (Power et al.; Nanda et al.), so your results can be compared with theirs.

Worked example: With $p = 113$ there are $113^2 = 12{,}769$ pairs. For $a = 100$, $b = 50$: $(100 + 50)$ mod $113 = 37$.

Self-check: How many possible answers does the model choose between?

Likely viva question: "Why $p = 113$ and not another prime?" (Answer: it matches Nanda et al. 2023. In the repo this is `task.modulus` in `06_Code/configs/nanda_unified.yaml`.)

2.2 Train/test split (30%)

What it is: 30% of the pairs are used for training and the other 70% are held back to test generalisation.

Why it matters: Grok is measured on the test set, so the split must be fixed per seed and never leak. In your code the 30% is hard-coded in `get_dataloaders` (`06_Code/src/data/modular_arithmetic.py`: `train_size = int(0.3 * len(...))`); the config's `train_fraction` does not change that split.

Worked example: $0.3 \times 12{,}769 \approx 3{,}830$ training pairs and $\approx 8{,}939$ test pairs. Exactly, $\lfloor 0.3 \times 12{,}769 \rfloor = 3{,}830$ and $12{,}769 - 3{,}830 = 8{,}939$.

Self-check: What goes wrong if a training pair appears in the test set?

Likely viva question: "How do you guarantee the test set is truly unseen, and that it's the same split when you reload a checkpoint?"

2.3 Transformer, logits and cross-entropy

What it is: A one-layer, four-head transformer reads the two numbers and outputs 113 scores (logits), one per possible answer; cross-entropy loss punishes low probability on the correct answer.

Why it matters: Every predictor is computed from this shared model's weights or outputs.

Worked example: If the softmax gives the correct answer probability 0.25, loss $= -\ln 0.25 \approx 1.39$; at 0.99 it is ≈ 0.01.

Formula: $\text{softmax}(z)_k = e^{z_k} / \sum_j e^{z_j}$, loss $= -\ln \text{softmax}(z)_{\text{correct}}$.

Self-check: Why does adding the same constant to all 113 logits not change the loss?

Likely viva question: "Is your model exactly Nanda's?" (Known gap: no attention output matrix W_O.)

2.4 AdamW and weight decay

What it is: AdamW is the optimiser that updates the weights; weight decay shrinks every weight a little each step, pulling the model towards smaller weights.

Why it matters: You use $\lambda = 1.0$ with betas $(0.9, 0.98)$, as in Nanda (`optimizer` in `nanda_unified.yaml`). Weight decay is believed to drive grokking, and it can also make weight-based signals fall on their own.

Worked example: Decoupled weight decay roughly does $w \leftarrow w - \eta\lambda w$ each step, on top of the gradient step.

Self-check: Why could a falling weight norm be "caused by the optimiser" rather than "a sign of grokking"?

Likely viva question: "Did you vary λ? If not, how can you separate your signal from the effect of weight decay?"

2.5 Epochs and checkpoints

What it is: An epoch is one full pass over the training data; a checkpoint is a saved copy of the weights at a given epoch.

Why it matters: All runs last 40,000 epochs. Events and grok are reported in epochs, and several predictors are computed later from checkpoints.

Worked example: The code saves a checkpoint every 50 epochs, plus the final epoch (`SAVE_EVERY` in `06_Code/scripts/run_nanda_benchmark.py`). Predictors are evaluated on the saved epochs that fall on the default grid of every 100 epochs (`EVAL_EVERY`). An event on that grid can only be located to within 100 epochs.

Self-check: Why must the data split be rebuilt with the same seed when re-reading a checkpoint?

Likely viva question: "What's your time resolution, and could it change any verdict?"

2.6 Seeds and why five

What it is: A seed fixes the random numbers (initial weights, data split, batch order), so each seed is one independent, repeatable run.

Why it matters: The frozen protocol scores 5 seeds (`evaluate_frozen_protocol` takes length-5 lists). Different seeds grok at different times, and that spread is what lets you test whether a predictor tracks *when* grok happens. `nanda_unified.yaml` only sets `per_predictor_min_runs: 3`; the scored runs are the five seeds below.

Worked example: Grok epochs for seeds 0–4: 12,987, 6,921, 9,510, 11,019, 23,747 (mean about 12,836).

Self-check: Why is one seed not enough to judge a predictor?

Likely viva question: "Five is very few. What can and can't you conclude from five seeds?"

Module 3: Measuring the outcome (2 topics)

3.1 Test accuracy

What it is: The fraction of test pairs the model answers correctly.

Why it matters: It's the only thing that defines grok in your benchmark.

Worked example: 8,100 correct out of 8,939 test pairs gives ≈ 0.906.

Self-check: Why not use test loss instead?

Likely viva question: "Why accuracy and not loss to define grokking?"

3.2 The grok epoch G

What it is: The first epoch where test accuracy goes above 0.9.

Why it matters: It's the target every predictor is compared with. Recomputing it from saved curves reproduces your saved values. The primary threshold is the first entry of `evaluation.grok_thresholds` in `nanda_unified.yaml`, which is 0.9.

Worked example: $G = (12987, 6921, 9510, 11019, 23747)$ for seeds 0–4.

Definition: $G = \min\{t : \text{acc}_{\text{test}}(t) > 0.9\}$.

Self-check: Would G change much with a 0.95 threshold? Why does it matter that the threshold is fixed?

Likely viva question: "Why 0.9? Is your result sensitive to that choice?"

Module 4: What a predictor is (5 topics)

4.1 Predictor and event epoch P

What it is: A predictor is a quantity tracked during training plus a rule that names one epoch P (the "event") as the warning.

Why it matters: Each of your eight predictors is a (signal, rule) pair. If the rule changes, it's a different predictor.

Worked example: L2's crossover rule gives $P = (113, 113, 120, 131, 137)$ (more precisely 113.4, 113.2, 120.1, 130.6, 136.8).

Self-check: What should the code do if the rule never fires (a "miss", `None`)? See 4.5 for the rule that is actually coded.

Likely viva question: "Which exact rule is the official one for this predictor?"

4.2 Lead and the 100-epoch margin

What it is: A firing seed counts as "led" only if the event comes at least 100 epochs before grok: $P < G - 100$.

Why it matters: The margin stops an event that fires at grok from counting as a warning. A miss (`None`) is not a lead. It is dropped before this comparison (see 4.5).

Worked example: Seed 0: $113 < 12987 - 100 = 12887$, so it leads. An event at $P = 12950$ would not.

Self-check: Does the summary table's "before grok" column use the same margin? (No: audit finding, it uses > 0. That column is `seeds_before_grok` in `summarise`, which counts $G - P > 0$, not the 100-epoch margin.)

Likely viva question: "Why 100 epochs?"

4.3 Lead fraction

What it is: The share of *firing* seeds that lead. When every seed fires, lead_frac $= \#\{P < G - 100\}/5$.

Why it matters: PASS needs lead_frac ≥ 0.8. With all five seeds firing, that is at least 4 of 5. Misses change the denominator; see 4.5.

Worked example: L2 crossover: all 5 lead, so lead_frac $= 5/5 = 1.0$.

Self-check: If one seed misses and one of the other four fires after grok, what is `lead_frac`? (Four seeds fired, three of them lead, so $3/4 = 0.75$, which fails ≥ 0.8. It is not $3/5$.)

Likely viva question: "Is 4 out of 5 strong evidence?"

4.4 Relative lead and mean relative lead

What it is: How early the event is, as a fraction of grok time: $r = (G - P)/G$. The mean $\bar{r}$ is taken over the seeds that fired.

Why it matters: PASS needs $0.05 \leq \bar{r} \leq 0.9$. Too small means "too late to be useful"; too large means the event is so early it's probably unrelated to grok.

Worked example: Seed 0 of L2: $r = (12987 - 113)/12987 \approx 0.991$. Mean over five seeds $\bar{r} \approx 0.989 > 0.9$, so it fails this part.

Self-check: What $\bar{r}$ does an event exactly halfway to grok give?

Likely viva question: "Why have an upper limit at all?"

4.5 Misses and `None`

What it is: A miss is a seed where the rule never fires. The code stores that as `None`.

Why it matters: `evaluate_frozen_protocol` in `06_Code/scripts/analyze_nanda_unified.py` handles `None` like this. It drops `None` pairs before Spearman's ρ and before the mean relative lead. If fewer than 4 of the 5 seeds fire, it fails at once and returns lead_frac $= 0$, $\rho =$ NaN, and mean relative lead $= 0$. If 4 or 5 seeds fire, lead_frac, ρ, and $\bar{r}$ are computed only on those firing seeds. A miss is not put in as a failed lead inside that average.

Worked example: One miss and four leads: four seeds fired, so this is not the immediate fail. lead_frac $= 4/4 = 1.0$, and ρ is Spearman on those four pairs.

Self-check: Can you compute ρ with 4 seeds? Yes, when exactly four seeds fire. With three or fewer, ρ is not computed.

Likely viva question: "How do you treat a seed where the signal never appears?"

Module 5: Spearman's ρ and small samples (4 topics)

5.1 Spearman rank correlation ρ

What it is: A number from -1 to 1 saying whether two lists go up together *in order*, using only ranks. (Rank: 1 = smallest, 2 = next, and so on.)

Why it matters: PASS needs $\rho(P, G) \geq 0.9$: seeds that grok later must also have later events. This shows the event tracks *this run's* grok, not a fixed clock.

Formula (no ties): $\rho = 1 - \dfrac{6 \sum d_i^2}{n(n^2-1)}$, where d_i is the rank difference for seed i. With ties, compute Pearson's r on the (averaged) ranks. Pearson's r is the ordinary straight-line correlation; it is defined in 5.2.

Worked example (L2 crossover): Grok ranks: $G = (12987, 6921, 9510, 11019, 23747) \to (4, 1, 2, 3, 5)$. Event ranks: $P = (113.4, 113.2, 120.1, 130.6, 136.8) \to (2, 1, 3, 4, 5)$. $d = (2, 0, -1, -1, 0)$, $\sum d^2 = 6$, $\rho = 1 - \dfrac{6 \cdot 6}{5 \cdot 24} = 1 - 0.3 = 0.70$. Fails ($< 0.9$).

If you round the events to $(113, 113, 120, 131, 137)$ the first two tie at rank 1.5, and ρ (Pearson on ranks) becomes $5.5/\sqrt{10 \cdot 9.5} \approx 0.56$. Rounding changes the answer, which is why ties matter.

Self-check: What ρ does an event at a fixed epoch (same in all seeds) get?

Likely viva question: "Why Spearman and not Pearson?"

5.2 Pearson vs Spearman

What it is: Pearson measures a straight-line relationship using raw values; Spearman only uses order.

Why it matters: With 5 seeds, seed 4 ($G = 23747$) is far from the rest and dominates Pearson. Spearman is not pulled by one extreme value.

Worked example: Any event that rises with G in exactly the same order has $\rho = 1$, even if the relationship is curved.

Self-check: Which one would seed 4 distort, and why?

Likely viva question: "Your Pearson values look strong; why don't you trust them?"

5.3 Only 21 possible values with 5 seeds

What it is: With $n = 5$ and no ties, $\rho = 1 - \sum d^2/20$, and $\sum d^2$ is an even number from 0 to 40, so ρ can only be $-1, -0.9, \ldots, 0.9, 1$: 21 values.

Why it matters: $\rho \geq 0.9$ means "perfect order, or at most one adjacent pair swapped". It also explains why the code uses a cutoff just below 0.9 (`RHO_THRESH = 0.8999999999999998` in `analyze_nanda_unified.py`) to absorb floating-point rounding.

Worked example: One swap of neighbours gives $\sum d^2 = 2$, $\rho = 0.9$. Two swaps gives 0.8, fail.

Self-check: Can $\rho = 0.85$ happen with 5 seeds and no ties?

Likely viva question: "Isn't the ρ test very coarse with five seeds?"

5.4 Significance with small n

What it is: Whether a correlation could easily appear by chance. (A p-value is the usual name for that chance; your protocol does not use one.)

Why it matters: With $n = 5$, only $\rho = 1$ (and borderline 0.9) is unlikely by chance; $\rho = 0.70$ is not significant.

Worked example: Random ordering of 5 seeds gives $\rho = 1$ with probability $1/120$.

Self-check: Why does your protocol use a threshold rather than a p-value?

Likely viva question: "How confident can you be in a PASS from five points?"

Module 6: The frozen protocol and the PASS rule (3 topics)

6.1 The frozen protocol

What it is: A fixed set of rules (config `nanda_unified.yaml`, $p = 113$, 5 seeds, grok threshold 0.9, the PASS thresholds) written down and not changed while predictors are scored.

Why it matters: It makes all predictors comparable and stops thresholds being tuned to make a favourite predictor pass. It was written on 29 September 2026, after L2 had been closed, which is itself an attack point.

Worked example: The verdict function `evaluate_frozen_protocol` applies the three conditions below.

It lives in `analyze_nanda_unified.py`. Only signals listed in `PRIMARY_SIGNALS` in that file receive this verdict.

Self-check: Which predictors were scored by this function and which were only re-computed later?

Likely viva question: "Was the protocol fixed before you saw any results?"

6.2 The PASS rule

What it is: A predictor passes only if all three hold:

$$\text{lead_frac} \geq 0.8, \qquad \rho(P, G) \geq 0.9, \qquad 0.05 \leq \bar{r} \leq 0.9.$$

The code compares ρ with `RHO_THRESH` (just below 0.9), as in 5.3. The other two cuts are exact.

Why it matters: Each condition blocks a different failure: firing late, firing on a fixed clock, or firing absurdly early.

Worked example: L2 crossover: 1.0 (pass), 0.70 (fail), 0.989 (fail), so FAIL. L2 MA-of-MA: 1.0 (pass), 0.00 (fail), 0.835 (pass), so FAIL.

Self-check: Build a made-up event list that passes all three for your five G values.

Likely viva question: "Justify each threshold. Where do 0.8, 0.9, 0.05 and 0.9 come from?"

6.3 Pre-registration and criterion validation

What it is: Pre-registration means stating your test before running it; criterion validation means checking that the test can tell a good signal from a bad one, for example on synthetic data.

Why it matters: Your `context.md` records a calibration of the criterion. If it was done after some results were known, an examiner can call it post-hoc.

Worked example (illustrative, not a real run): These two rows are made-up controls, not measured predictors. A synthetic "perfect" predictor $P = 0.5\,G$ gives lead_frac 1, $\rho = 1$, $\bar{r} = 0.5$, so it must PASS. A fixed $P = 1800$ gives $\rho = 0$ and must FAIL.

Self-check: What would a synthetic control that *should* fail look like?

Likely viva question: "How do you know your criterion isn't too strict to pass anything?"

Module 7: Research honesty (6 topics)

7.1 Hindsight / look-ahead bias

What it is: Using information from the future (for example the final value of a curve) to decide when an earlier event happened.

Why it matters: Two rules ("L2 halfway down from peak", "Dropout at 10% of its rise") only pass because they use the run's final value. A live monitor couldn't do that.

Worked example: Target $= \text{peak} - 0.5(\text{peak} - y_{\text{final}})$ needs y_{final} at epoch 40,000.

Self-check: How would you test whether a rule is look-ahead free?

Likely viva question: "Could this signal be used during training, before you know the outcome?"

7.2 Causal vs centred moving average

What it is: A causal average at epoch t uses only epochs $\leq t$; a centred one uses epochs on both sides.

Why it matters: `uniform_filter1d` is centred, so smoothed curves "know" the future. Any event found on them is slightly look-ahead.

Worked example: Causal window 3: $(y_{t-2} + y_{t-1} + y_t)/3$. Centred window 3: $(y_{t-1} + y_t + y_{t+1})/3$.

Self-check: Which one shifts events later?

Likely viva question: "Is your smoothing causal?"

7.3 Researcher degrees of freedom / tuning to the data

What it is: Choices (windows, thresholds, skip epochs) made after looking at the results.

Why it matters: Several rules were changed after looking at curves; an examiner can say they were fitted to the data.

Worked example: L2's `skip_epochs` went from 200 to 100 after inspecting curves, and the crossover then fires just after 100. The current value is `l2_norm.skip_epochs: 100` in `nanda_unified.yaml`.

Self-check: Which settings in your config say "not from a paper"?

Likely viva question: "How many rules did you try before this one?"

7.4 Sensitivity analysis

What it is: Changing a setting a little and checking whether the verdict changes.

Why it matters: None of the window sizes or margins were varied. Near-misses show how fragile some verdicts are.

Worked example: The L2 peak event gives $\bar{r} = 0.904$, failing only by 0.004.

Self-check: Which threshold would you vary first?

Likely viva question: "Would your verdict survive a different window size?"

7.5 Baselines, controls and noise floor

What it is: A baseline or control run is one where grokking doesn't happen (or a setting is turned off). The noise floor is how much a signal wobbles for no reason.

Why it matters: Without them you can't say a change in a signal is *because of* grokking.

Worked example: A run with weight decay $\lambda = 0$ (which often doesn't grok) would show what $\sum w^2$ does without grokking.

Self-check: How would you estimate the noise floor from the saved curves?

Likely viva question: "Where's your control condition?"

7.6 Negative results and reproducibility

What it is: A negative result is a clear "this doesn't work"; reproducibility means someone else can rerun and get the same numbers.

Why it matters: Seven of eight predictors are closed-negative, so the thesis value is in showing the tests were fair and repeatable (fixed seeds, saved data, frozen thresholds).

Worked example: Recomputing from saved accuracy curves gives the same grok epochs; rebuilding the split reproduces saved accuracies.

Self-check: What makes a negative result publishable rather than just a failure?

Likely viva question: "If almost nothing passes, what is the contribution of the thesis?"

Recommended reading (short)

1. Power, A., Burda, Y., Edwards, H., Babuschkin, I., Misra, V. (2022). *Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets.* arXiv:2201.02177. (Modules 1–2.)

   Local file:

   `04_Literature/Papers/Alethea Power - Grokking Generalization beyond overfitting on small algorithmic datasets [2022].pdf`

2. Nanda, N., Chan, L., Lieberum, T., Smith, J., Steinhardt, J. (2023). *Progress Measures for Grokking via Mechanistic Interpretability.* ICLR 2023. (Modules 1–2; your setup.) Local file: `04_Literature/Papers/Nanda et al.pdf`.

3. Liu, Z., Michaud, E. J., Tegmark, M. (2023). *Omnigrok: Grokking Beyond Algorithmic Data.* ICLR 2023. (Weight norm and weight decay.) Not in `04_Literature/`.

4. Loshchilov, I., Hutter, F. (2019). *Decoupled Weight Decay Regularization.* ICLR 2019. (AdamW.) Not in `04_Literature/`.

5. Any intro statistics text chapter on Spearman rank correlation (for example the Spearman 1904 original, or a standard textbook). (Module 5.) Not in `04_Literature/`.

6. Nosek, B. A. et al. (2018). *The Preregistration Revolution.* PNAS 115(11). (Modules 6–7.) Not in `04_Literature/`.